**Clinical Study Summary Report (CSSR) Document Status:** Final | **Version:** 1.0 | **Date:** 06-Apr-2026

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## 1. Administrative & Study Identification

**Full Study Title:** Safety and Efficacy of Lenvatinib (E7080/ MK-7902) in Combination With Pembrolizumab (MK-3475) Versus Lenvatinib as First-line Therapy in Participants With Advanced Hepatocellular Carcinoma (MK-7902-002/E7080-G000-311/LEAP-002) **Official Title:** A Phase 3 Multicenter, Randomized, Double-blinded, Active-controlled, Clinical Study to Evaluate the Safety and Efficacy of Lenvatinib (E7080/MK-7902) in Combination With Pembrolizumab (MK-3475) Versus Lenvatinib in First-line Therapy of Participants With Advanced Hepatocellular Carcinoma (LEAP-002) **Short Title/Acronym:** LEAP-002 **Protocol Number:** MK-7902-002 / E7080-G000-311 **NCT Number:** NCT03713593 **Posting ID:** 450044681376952930 **Sponsor/Company Name:** Merck (Merck Sharp & Dohme LLC in collaboration with Eisai Inc.) **Investigational Product:** Lenvatinib (Tyrosine Kinase Inhibitor) + Pembrolizumab (Immune Checkpoint Inhibitor) **Phase of Development:** PHASE3 (Phase Requirement: Trial must be in PHASE3) **Trial Status:** COMPLETED (Study has been completed) **Medical Monitor:** Sponsor Clinical Operations Lead

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## 2. Study Synopsis & Rationale

**2.1 Study Objective & Summary Summary:** The purpose of this study is to evaluate the safety and efficacy of lenvatinib (E7080/ MK-7902) in combination with pembrolizumab (MK-3745) versus lenvatinib in combination with placebo as first-line therapy for the treatment of advanced hepatocellular carcinoma in adult participants. **Primary Hypotheses:** The primary hypotheses of this study are that lenvatinib plus pembrolizumab is superior to lenvatinib plus placebo with respect to progression-free survival (PFS) and overall survival (OS).

**2.2 Background & Rationale** Systemic therapies have improved the management of hepatocellular carcinoma (HCC), but there is still a significant need to further enhance overall survival in first-line advanced stages. HCC has a poor prognosis and limited initial treatment options. Building on promising clinical activity observed in earlier phase 1b studies, this trial evaluates the addition of pembrolizumab (anti-PD-1) to lenvatinib (multikinase inhibitor) to determine if the combination provides an added survival benefit and stronger anti-tumor activity compared to lenvatinib monotherapy.

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## 3. Study Design & Methodology

**3.1 Design Framework Study Type:** Interventional **Control Type:** Placebo-controlled, active comparator (Lenvatinib + Placebo) **Blinding:** Double-blinded **Randomization:** Randomized 1:1 ratio with stratification by geographical region, macrovascular portal vein invasion or extrahepatic spread, $\alpha$-fetoprotein concentration, and ECOG performance status.

**3.2 Planned Enrollment & Duration Targeted Enrollment ($N$):** 794 participants **Number of Sites:** 172 global sites **Estimated Study Start:** 31-Dec-2018 **Estimated Primary Completion:** 21-Jun-2022

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## 4. Subject Selection (Eligibility Criteria)

**4.1 Inclusion Criteria** 1. Is male or female and $\ge 18$ years of age at the time of signing the informed consent. 2. Has a diagnosis of hepatocellular carcinoma confirmed by radiology, histology, or cytology. 3. Has Barcelona Clinic Liver Cancer (BCLC) Stage C disease, or BCLC Stage B disease not amenable to locoregional therapy or refractory to locoregional therapy, and not amenable to a curative treatment approach. 4. Has a Child-Pugh class A liver score. 5. Has a predicted life expectancy of >3 months. 6. Has at least one measurable hepatocellular carcinoma (HCC) lesion based on RECIST 1.1 as confirmed by BICR. 7. Has an Eastern Cooperative Oncology Group (ECOG) performance status (PS) of 0 to 1. 8. Participants with hepatitis B will be eligible as long as their virus is well controlled.

**4.2 Exclusion Criteria** 1. Has had esophageal or gastric variceal bleeding within the last 6 months. 2. Has gastrointestinal malabsorption, gastrointestinal anastomosis, or any other condition that might affect the absorption of lenvatinib. 3. Has a preexisting Grade $\ge 3$ gastrointestinal or non-gastrointestinal fistula. 4. Has clinically significant hemoptysis from any source or tumor bleeding within 2 weeks prior to the first dose of study intervention. 5. Has significant cardiovascular impairment within 12 months of the first dose of study intervention such as history of congestive heart failure greater than New York Heart Association (NYHA) Class II, unstable angina, myocardial infarction or cerebrovascular accident stroke, or cardiac arrhythmia associated with hemodynamic instability. 6. Has had major surgery to the liver within 4 weeks prior to the first dose of study intervention. 7. Has had a minor surgery (ie, simple excision) within 7 days prior to the first dose of study intervention. 8. Has serious non-healing wound, ulcer, or bone fracture. 9. Has received any systemic chemotherapy for HCC or chemotherapy for any malignancy in the past 3 years. 10. Has received prior therapy with an anti-programmed cell death 1 (anti-PD-1), anti-programmed cell death ligand 1 (anti-PD-L1), or anti-programmed cell death ligand 2 (anti-PD-L2) agent or with an agent directed to another stimulatory or co-inhibitory T-cell receptor (eg, cytotoxic T-lymphocyte-associated protein 4 [CTLA-4], OX-40, or CD137). 11. Has a diagnosis of immunodeficiency or is receiving chronic systemic steroid therapy or any other form of immunosuppressive therapy within 7 days prior the first dose of study intervention. 12. Has a known additional malignancy that is progressing or has required active treatment within the past 3 years with the exceptions of basal cell carcinoma of the skin, squamous cell carcinoma of the skin, or carcinoma in situ (eg, breast carcinoma, cervical cancer in situ) that has undergone potentially curative therapy. 13. Has a known history of, or any evidence of, central nervous system (CNS) metastases and/or carcinomatous meningitis as assessed by local site investigator. 14. Has severe hypersensitivity ($\ge$ Grade 3) to study intervention and/or any of their excipients. 15. Has an active autoimmune disease that has required systemic treatment in past 2 years. 16. Has a history of (non-infectious) pneumonitis that required steroids or has current pneumonitis. 17. Has urine protein $\ge 1$ grams/24 hours. 18. Prolongation of corrected QT (QTc) interval to >480 milliseconds (corrected by Fridericia Formula). 19. Has left ventricular ejection fraction (LVEF) below the institutional normal range as determined by multigated acquisition scan (MUGA) or echocardiogram (ECHO). 20. Has an active infection requiring systemic therapy with the exceptions of hepatitis B virus (HBV) or hepatitis C virus (HCV); known history of human immunodeficiency virus (HIV) infection; or known active tuberculosis (Bacillus tuberculosis). 21. Has a history or current evidence of any condition, therapy, or laboratory abnormality that might confound the results of the study, interfere with the participant's participation for the full duration of the study, or is not in the best interest of the participant to participate, in the opinion of the treating investigator. 22. Has a known psychiatric or substance abuse disorder that would interfere with the participant's ability to cooperate with the requirements of the study. 23. Is pregnant or breastfeeding or expecting to conceive or father children within the projected duration of the study, starting with the screening visit through 120 days after the last dose of study intervention. 24. Has had an allogenic tissue/solid organ transplant.

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## 5. Intervention & Treatment Plan

**5.1 Investigational Regimen Investigational Product 1:** * **Drug Name:** lenvatinib * **Trade Name:** Lenvima * **ATC Code:** L01EX08 * **EMA Reference:** [EMA Product Information] * **Dosage/Frequency:** 12 mg orally once daily (for body weight $\ge 60$ kg) OR 8 mg orally once daily (for body weight $< 60$ kg). Continuous until progressive disease or unacceptable toxicity.

**Investigational Product 2:** * **Drug Name:** Pembrolizumab (MK-3475) * **Dosage/Frequency:** 200 mg on Day 1 of each 3-week cycle. IV Infusion. * **Duration of Treatment:** Pembrolizumab for up to 35 cycles (approx. 2 years) or until progressive disease/ unacceptable toxicity.

**5.2 Concomitant & Prohibited Therapy Permitted:** Standard supportive care for cancer-related symptoms and management of treatment-related toxicities. **Prohibited:** Concurrent investigational agents, other systemic anti-cancer therapies for HCC, and live vaccines.

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## 6. Study Timeline & Visit Schedule

| Visit ID | Window (Days) | Key Activities/Procedures |
| :--- | :--- | :--- |
| **Screening** | Day -28 to 0 | Informed Consent, Medical History, Baseline Imaging, Laboratory Assessments |
| **Baseline** | Day 1 | Randomization, First Dose (IV and Oral), Vital Signs |
| **Treatment Cycles** | Every 3 Weeks (Q3W) | IV Infusion (Pembrolizumab/Placebo), Safety Labs, Dispense Lenvatinib |
| **Tumor Imaging** | Every 9 Weeks | Radiographic tumor assessments (CT/MRI) evaluated by BICR |
| **End of Study** | Follow-up | Survival tracking, subsequent anti-cancer therapy monitoring |

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## 7. Outcome Measures (Endpoints)

**7.1 Primary Outcomes** 1. **Progression-free Survival (PFS) Per Response Evaluation Criteria in Solid Tumors Version 1.1 (RECIST 1.1):** PFS was defined as the time from the date of the first documentation of disease progression, as determined by blinded independent central review (BICR) per RECIST 1.1 or death due to any cause (whichever occurred first). Disease progression was defined as at least 20 percent (%) increase (including an absolute increase of at least 5 millimeter [mm]) in the sum of diameter of target lesions, taking as reference the smallest sum and/or unequivocal progression of existing non-target lesions and/or appearance of 1 or more new lesions. PFS was estimated and analyzed using Kaplan-Meier method. 2. **Overall Survival (OS):** OS was defined as the time from randomization until death from any cause.

**7.2 Secondary Outcomes** 1. **Objective Response Rate (ORR)** Per Response Evaluation Criteria in Solid Tumors Version 1.1 (RECIST 1.1). 2. **Duration of Response (DOR)** Per Response Evaluation Criteria in Solid Tumors Version 1.1 (RECIST 1.1). 3. **Disease Control Rate (DCR)** Per Response Evaluation Criteria in Solid Tumors Version 1.1 (RECIST 1.1). 4. **Safety and tolerability profiles.**

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## 8. Safety & Adverse Event Reporting

**Adverse Event (AE) Monitoring:** Continuous monitoring of Treatment-Related Adverse Events (TRAEs), with specific focus on immune-mediated AEs (e.g., hypothyroidism) and TKI-mediated AEs (e.g., hypertension, diarrhea, proteinuria). **Serious Adverse Events (SAEs):** Monitored throughout the study and for up to 120 days after the last dose. **Data Monitoring Committee (DMC):** Independent Data Monitoring Committee to evaluate safety and efficacy at interim and final analyses.

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## 9. Statistical Analysis Plan (SAP) Summary

**Analysis Populations:** Intention-to-Treat (ITT) population evaluated for all primary efficacy endpoints. **Sample Size Justification:** Designed for 794 patients to meet dual primary endpoints with pre-specified superiority boundaries. **Handling of Missing Data:** PFS was estimated and analyzed using Kaplan-Meier method. Stratified log-rank tests applied.

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## 10. Quality Assurance & Regulatory Compliance

**GCP Compliance:** This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Central interactive voice-response system utilized for randomization. **Audit Trail:** Radiographic images strictly audited and analyzed via Blinded Independent Central Review (BICR).

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## 11. Study Identifiers & Governance

**Primary ID:** {{MK-7902-002 / E7080-G000-311}} **Posting ID:** {{450044681376952930}} **Registry Number:** {{NCT03713593}} **Sponsor/ Lead Organization:** {{Merck}} **Study Title:** {{LEAP-002}} **Trial Status:** {{COMPLETED}}

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## 12. Clinical Framework (Hepatocellular Carcinoma Specific)

**Primary Condition:** {{Carcinoma, Hepatocellular}} **Disease Name/ Preferred Name:** {{Carcinoma}} **Preferred UMLS Name:** {{Hepatocellular Carcinoma, Drug Therapy}} **Semantic Type:** {{Neoplastic Process}} **Definition:** {{A malignant tumor arising from epithelial cells. Carcinomas that arise from glandular epithelium are called adenocarcinomas, those that arise from squamous epithelium are called squamous cell carcinomas, and those that arise from transitional epithelium are called transitional cell carcinomas (NCI Thesaurus).}} **Classification Codes:** * **MeSH Code:** {{D002277}} * **HPO Code:** {{HP:0030731}} * **SNOMED ID:** {{1187425009}} **Diagnosis Threshold:** {{BCLC Stage B or C; Child-Pugh class A}} **Medical Methodology:** {{Systemic Therapy: Tyrosine Kinase Inhibitor + Immune Checkpoint Inhibitor}}

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## 13. Study Procedures & Methodology

**Management Pathway:** {{First-line systemic combination therapy}} **Education Protocol (HCP):** {{Management of immune-mediated AEs and TKI-specific toxicities}} **Education Protocol (Patient):** {{Oral adherence and symptom reporting}} **Quality Audit Mechanism:** {{Blinded independent central review (BICR) of imaging}}

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## 14. Observation & Follow-Up Schedule

**Total Duration:** {{Up to 35 cycles (approx. 2 years) for IV; until progression for oral lenvatinib}} **Onsite Visit Cadence:** {{Every 3 weeks (Q3W)}} **Remote Monitoring Frequency:** {{Continuous AE and survival tracking}} **Checklist Review Interval:** {{Imaging assessments every 9 weeks}}

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## 15. Sign-off & Approval

| Role | Name | Signature | Date | | :--- | :--- | :--- | :--- | | **Principal Investigator** | [Name] | ____ **| [DD-MMM-YYYY] | | Clinical Operations Lead | [Name] |** ___ **|** ***[DD-MMM-YYYY]* | | *Lead Statistician* | *[Name]* |** _____ | [DD-MMM-YYYY] |